

Policy Memo Assignment

PP5182 Economic Diplomacy in the Age of Strategic Interdependence

To: Malaysia's Ministry of Investment, Trade and Industry (MITI)

From: Siyeon (Jenny) Kim

Subject: Enhancing Malaysia's domestic semiconductor industry to respond to US or China's export controls and tariffs

Date: 18th September 2026

1. Summary

Malaysia's advancement into a high-tech nation may be postponed due to US-China's frequently changing policies towards semiconductor partner countries. It is time for Malaysia to develop its domestic capabilities and industries in the long run. It is also important to create tools to monitor any policy changes from US and China to inform businesses of any protocol changes and recommend advice to continue stable business operations. Measures to counter short-term effects of these policy decisions as well as ways to develop domestic industries that will secure Malaysia's economic growth in the long run will be vital.

2. Background

Malaysia has transitioned from being an agricultural-based economy into an industrial economy since the 1970s when the manufacturing industry first began to be developed. The [World Bank's data](#) shows that this industry's value-added contribution to Malaysia's economy increased from 14% in 1970 to 22% in 2025. As such, after five decades, the manufacturing industry has become the [second largest](#) business sector in Malaysia and generates one quarter of Malaysia's value-added GDP. While the Electrical and Electronics (E&E) sector was highly contributive to this development, the semiconductor industry was also developed from the 1970s and has successfully integrated into the global supply chain of semiconductors. [13% of global semiconductor](#) back-end assembly processing takes place in data facilities located in Malaysia. This suggests that Malaysia's semiconductor industry is highly competitive among South-east Asian countries and is reliable for larger partner countries.

However, Malaysia is reluctant to remain in the lower-end of the supply chain and seeks to progress into higher-end manufacturing stages, as it will make the economy more productive.

The [New Industrial Master Plan 2030](#) outlines Malaysia's vision to enhance its competitiveness by increasing economic complexity, the number of high-value jobs, and connectivity between domestic and foreign firms. The action points of strategy 1.1 state that creating IC design businesses and establishing wafer fabrication facilities domestically will be important. It is well known that the semiconductor industry's development will be critical for Malaysia's economy.

Currently, in Malaysia, foreign investment plays a crucial role in developing the semiconductor industries. However, attracting funding from US or Chinese companies may be challenging as they have already established manufacturing centres in states including Penang, Johor, and Negeri Sembilan, and the assembly, testing, and packaging (ATP) market is already competitive. Convincing these foreign companies to invest in business operations that require more expertise would mean attracting more skilled workers and providing a conducive business environment to support its development.

More importantly, the US and Chinese governments have been opposed to open trade of semiconductor supplies due to the potential military and security reasons. In 2022, the US government imposed export restrictions on advanced Nvidia chips, such as A100 and H100, by requiring Nvidia to obtain a US licence. Their rationale was to limit China access from obtaining them to support their potential development of military weapons. In the same year, the US Department of Commerce Bureau of Industry and Security (BIS) launched [broader export control](#) measures and has continued to tighten controls in 2025 on countries that may linked to transshipment routes with China. In response, Malaysia has requested companies to [notify exports or transshipment](#) of AI chips of US-origin not on the Strategic Items List (SIL) to issue a trade permit. More policies may appear constantly, which may inhibit the usual processes of trading semiconductor components with China. Moreover, the US may be reluctant to set up critical infrastructure in Malaysia that may allow the industry to develop advanced chips and higher-end products.

China has responded by imposing broad export controls on gallium and germanium, which are important resources for semiconductors, on Japan and the US previously in 2023. However, this policy has still affected trade today as [China's export of gallium](#) stood at merely 3kg in April 2026 which all went to Malaysia. China exported 4,777 kg of the same

metal to other countries in April 2025. As such, these cases show that Malaysia's trade will be affected by China and US's constantly changing export controls and tariff policies.

3. Analysis of Options

First, in order to defend Malaysia's semiconductor industry's development, solely relying on the US or China's supply of materials might cause uncertainty and hinder smooth business operations. As such, Malaysia needs to build its domestic semiconductor industry by improving the current level of technical expertise and production capabilities. It would require investing in STEM education and spending more GDP into R&D in academic institutions and firms. According to [MASTIC](#), 1.01% of GDP was spent on Malaysia's R&D in 2022. More research opportunities between universities and semiconductor companies should be created to allow local products to be invented in the future. Developing Malaysia's local industry would be beneficial as Malaysia's private or public firms have not issued many patents compared to South Korea or Taiwan, which was a critique raised by a [publication from IAEAS](#). However, this strategy would require a long-term approach. Therefore, enhancing research roles and building higher educational standards will have a positive long-term effect on Malaysia's semiconductor development.

Second, Malaysia should build a more competitive environment for global companies to invest in higher-tech developments. Malaysia is a politically stable, relatively international country with low costs of living and sufficient skilled talent. However, most foreign investment is focused on the lower-end of the semiconductor supply chain process, so Malaysia would need to provide favorable economic conditions for US or Chinese companies seeking to build higher technology facilities in Malaysia.

Lastly, Malaysia should continue to monitor any new tariffs or export policies implemented by the US' or China's governments. Furthermore, if China seeks to increase its businesses in the country, Malaysia should create a regulation to allow other states to be invested rather than solely in Penang, which already has enough manufacturing factories. This is to protect its companies from receiving external policies or economic threats from the US or China that can impact its semiconductor industry.

4. Recommendation

Malaysia needs to improve its domestic semiconductor industry although it may mean reducing the speed of progress due to higher costs on companies. This is to ensure that Malaysia has sufficient talent, technological capabilities, and supply chains to develop a strong self-sufficient industry. It will also require bringing in investments from international companies other than the US and China and also allowing local talent to gain technological expertise from these companies. Moreover, improving students' learning on STEM subjects and improving the education system will be beneficial in building future talent and leaders in the semiconductor or other technological industries. These are long-term plans that Malaysia can take.

In the short term, Malaysia will need to constantly monitor any policy changes of the US or Chinese government to balance the needs of both parties. If there are conflicting interests, Malaysia may need to prioritize their relations with foreign companies that have more expertise in IC design and wafer fabrication technology to align their future semiconductor goals with these measures.

(Word count: 1196)

AI Declaration: I used ChatGPT Education (GPT-5.6 Sol) to research articles and find more points about Malaysia's semiconductor industry and US-China tensions. However, I conducted my own research, used resources to analyze Malaysia's current industry, and verified the sources given by ChatGPT. I am responsible for the content and quality of the submitted work.